HOUD LAOUAOUDJA

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Professional Summary

Results-driven 3rd-year CS student at ESI Algeria specializing in full-stack engineering, AI-driven IoT systems, and edge computing. Led a 6-person offline-first agricultural automation platform from concept to deployment, combining embedded systems, multi-agent AI, and mobile development. Seeking a software engineering internship focused on building scalable, real-world distributed systems.

Education

École Nationale Supérieure d'Informatique (ESI) — Algiers, Algeria
Superior Cycle / 3rd Year in Computer Science — 2024 – Present

- **Academic Ranking:** Ranked **20th / 344** (Top 6%) in 2nd Year Prep (2CP) • **16th / 429** (Top 4%) after 1st Year (1CP).
- **Coursework:** Data Structures (ALSDD), File Structures (SFSD), OOP (Java), Systems Architecture, Probability & Statistics.

Technical Skills

- **Languages:** Python, JavaScript, Java, C, SQL
- **Frontend:** React, React Native, HTML/CSS
- **Backend:** Node.js, Express.js, REST APIs
- **IoT & Embedded:** ESP32, Arduino, MicroPython, MQTT
- **Cloud & DevOps:** Firebase, Supabase, Git, GitHub
- **Automation & AI:** Home Assistant, n8n, Multi-Agent AI
- **Data Structures:** Trees, B-Trees, Linked Lists, Queues

Engineering Projects

SAHLA Farm – Smart Agriculture IoT Ecosystem — 2026
React Native • Home Assistant • n8n • AI Agents • ESP32 • MQTT

- Collaborated in a high-performing 6-member team to build an offline-first, edge-computing ecosystem (**SAHLA Box**) to ensure smart irrigation functionality in remote areas with unstable network connectivity.
- Programmed ESP32 controllers to track soil telemetry and establish secure, asynchronous communication feeds via the MQTT protocol directly to a centralized hardware hub.
- Designed multi-agent AI pipelines with custom confidence scoring to eliminate deterministic model hallucinations and successfully resolve low-light visual ambiguities.
- Automated conditional execution workflows using n8n, ensuring robust mobile data state synchronization cleanly into a responsive React Native mobile user app.

8086 Assembly Address Book System — 2024 – 2025
8086 Assembly • Low-Level Programming • Memory Management

- Developed a low-level address book in 8086 Assembly using segmented data structures to record, track, and align structured persistent entries directly inside RAM.
- Programmed low-level string index routines and modular parsing parameters to support full data model CRUD operations within strictly fixed memory boundaries.
- Engineered data validation logic directly across hardware assembly registers to optimize instruction execution, prevent buffer overflows, and enforce complete data integrity.

Sequential Data Structures – Q&A Game Engine — 2024
C • Linked Lists • Modular Design • ALSDD

- Implemented a terminal-based game engine in C engineered on dynamic linked-list structures to maintain sequential data workflows and flexible runtime memory footprints.
- Enforced modular decomposition by separating logic architectures across independent modules covering player state pipelines, sorting, and statistical calculations.

Personal Portfolio Website & Web Utilities — 2024
React • HTML/CSS • JavaScript

- Built a fully responsive personal portfolio workspace and state-managed utilities leveraging React component architectures to highlight technical projects and core web development skills.

Activities & Languages

Activities & Clubs

- Marketing Dept. – School of AI, ESI
- Developer Dept. – SEC Club, ESI

Languages

- Arabic (Native)
- English & French (Intermediate)